

SOP-001: Synthesis of High-Stability Solid-State Battery Electrolyte

Target Molecule: CS(=O)(=O)OCS(=O)(=O)OCS(=O)(=O)O (Sulfonate Ester Chain)
Origin: Shodh AI Engine 4 (Autonomous Retrosynthesis)

Safety & PPE

Hazards: Methanesulfonyl chloride is a severe lachrymator and corrosive. Sulfuric acid is highly corrosive and exothermically reactive.

PPE Required: Chemical-resistant gloves, splash goggles, and fume hood are mandatory.

Reagents & Volumes

Reagent / Material	Volume / Amount	Notes
Standard solvent	300.0 uL total	100.0 uL per reaction vial
Methanesulfonyl chloride	117.4 uL total	54.5 uL + 62.9 uL
Methanol	65.5 uL	CO proxy
Sulfuric acid (H2SO4)	53.3 uL	Final condensation reagent

Step-by-Step Bench Procedure

Step 1: Initial Sulfonyl Condensation

- 1 Prepare Reaction Vial 1. Add 100.0 uL of standard solvent.
- 2 Using a micropipette, carefully add 54.5 uL of methanesulfonyl chloride.
- 3 Add 65.5 uL of methanol.
- 4 Agitate/mix thoroughly, simulating 5 robotic pipette aspirations.
- 5 Let the reaction stir at room temperature for exactly 10 minutes.

Step 2: First Chain Extension

- 1 Prepare a clean Reaction Vial 2. Add 100.0 uL of standard solvent.
- 2 Extract exactly 80.0 uL of the crude intermediate from Vial 1 and transfer it into Vial 2.
- 3 Add an additional 62.9 uL of methanesulfonyl chloride to Vial 2.
- 4 Agitate/mix 5 times.
- 5 Let the reaction stir for 10 minutes.

Step 3: Final Chain Extension

- 1 Prepare a clean Reaction Vial 3. Add 100.0 uL of standard solvent.
- 2 Extract exactly 80.0 uL of the crude intermediate from Vial 2 and transfer it into Vial 3.
- 3 Add 53.3 uL of sulfuric acid to drive the final condensation.
- 4 Agitate/mix 5 times.
- 5 Let the reaction stir for 10 minutes.

Quality Control

Quality Control
Take the final crude product from Vial 3 and perform LCMS (Liquid Chromatography-Mass Spectrometry).
Target Mass: Verify the molecular weight corresponds to the target sulfonate ester.